

3.9. Building a Multi-Step Form with Oracle JET and Knockout.js with more advanced features

Introduction

This activity guides you through the creation of a dynamic, multi-step form using the Oracle JET framework and Knockout.js. You'll learn how to manage form data across different steps, implement client-side validation, handle file uploads, and display submitted data in a table. This practical exercise will solidify your understanding of reactive programming with Knockout.js within an Oracle JET context, demonstrating how to build a robust and user-friendly data entry experience.

Objectives

Upon completing this activity, you will be able to:

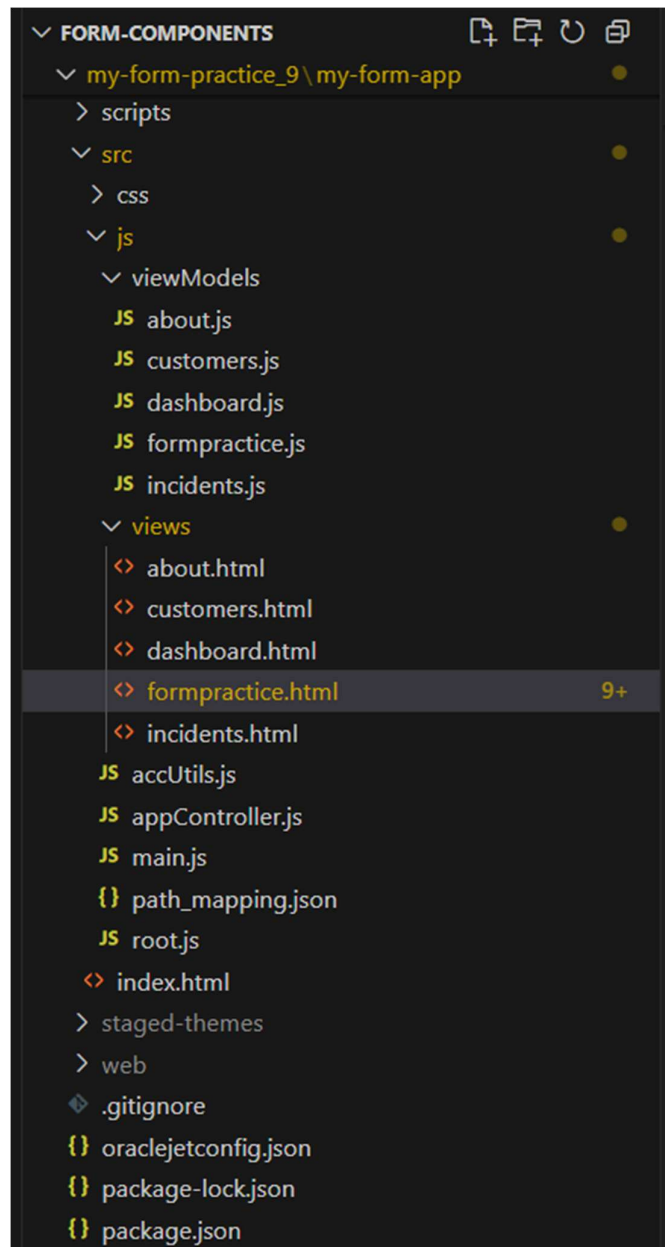
- Implement a multi-step form flow using Oracle JET's oj-train component.
- Manage form state across multiple steps using Knockout.js observables.
- Perform client-side form validation on individual steps.
- Dynamically populate dropdowns based on selections (e.g., states based on country).
- Integrate oj-c-file-picker for file upload functionality.
- Display submitted form data in an oj-table component.
- Generate and export form data as a JSON file.
- Utilize oj-messages and oj-dialog for user feedback and confirmations.
- Create derived observables for real-time UI updates, such as avatar initials.

Prerequisites

Before starting this activity, ensure you have:

- A basic understanding of HTML, CSS, and JavaScript.
- Familiarity with Oracle JET framework fundamentals.
- Working knowledge of Knockout.js concepts, including observables, computed observables, and data binding.
- Node.js and npm installed on your system.
- Oracle JET command-line interface (CLI) installed globally (npm install -g @oracle/ojet-cli).
- An Integrated Development Environment (IDE) such as Visual Studio Code, configured for web development.

Project structure:



1. Create a new folder named "my-form-app-practice_9" and move to that directory
2. In the command line run,
Ojet create my-form-app --template=navdrawer
3. Then run **ojet build web** and **ojet serve** respectively (like we did before)
4. Create folders `/js/viewModels/formpractice.js` and `/js/views/formpractice.html`

5. Add code to formpractice.js

```
define([
  'knockout',
  'ojs/ojarraydataprovider',
  'ojs/ojformlayout',
  'ojs/ojinputtext',
  'ojs/ojselectsingle',
  'ojs/ojradio',
  'ojs/ojcheckbox',
  'ojs/ojbutton',
  'ojs/ojdialog',
  'ojs/ojmessages',
  'ojs/ojtrain',
  'ojs/ojavatar',
  'ojs/ojlabel',
  'oj-c/file-picker',
  'ojs/ojtoolbar',
  'ojs/ojprogressbar',
  'ojs/ojlabelvalue',
  'ojs/ojtable',
  'jquery',
  // Ensure jQuery is correctly imported
], function (ko, ArrayDataProvider, ojtable) { // And mapped to $
  function FormPracticeViewModel() {
    this.trainSteps = [
      { id: 'step1', label: 'Personal Information' },
      { id: 'step2', label: 'User Preferences' },
      { id: 'step3', label: 'Review and Submit' },
      { id: 'step4', label: 'User Directory' }
    ];
    this.stepValue = ko.observable('step1');

    this.name = ko.observable('');
    this.email = ko.observable('');
    this.selectedCountry = ko.observable('');
    this.selectedState = ko.observable('');
    this.step1Messages = ko.observableArray([]);

    this.gender = ko.observable('');
    this.hobbies = ko.observableArray([]);
    this.profilePhoto = ko.observableArray([]);

    this.acceptArr = ko.observableArray([]);
    this.multipleStr = ko.observable('single');
    this.isDisabled = ko.observable(false);
  }
});
```

```

this.countryOptions = [
  { value: 'IN', label: 'India' },
  { value: 'US', label: 'USA' },
  { value: 'UK', label: 'UK' }
];
this.countryDataProvider = new ArrayDataProvider(this.countryOptions, {
  keyAttributes: 'value' });

this.stateMap = {
  IN: [
    { value: 'PB', label: 'Punjab' },
    { value: 'DL', label: 'Delhi' },
    { value: 'MH', label: 'Maharashtra' }
  ],
  US: [
    { value: 'NY', label: 'New York' },
    { value: 'CA', label: 'California' },
    { value: 'TX', label: 'Texas' }
  ],
  UK: [
    { value: 'ENG', label: 'England' },
    { value: 'SCT', label: 'Scotland' },
    { value: 'WLS', label: 'Wales' }
  ]
};

this.stateDataProvider = ko.pureComputed(() => {
  const country = this.selectedCountry();
  const states = this.stateMap[country] || [];
  return new ArrayDataProvider(states, { keyAttributes: 'value' });
});

this.isStateEnabled = ko.computed(() => !!this.selectedCountry());

this.avatarInitials = ko.computed(() => {
  const name = this.name();
  if (!name) return '';
  const parts = name.trim().split(' ');
  if (parts.length === 1) return parts[0].charAt(0).toUpperCase();
  return (parts[0].charAt(0) + parts[parts.length -
1].charAt(0)).toUpperCase();
});

this.formSubmitted = ko.observable(false);
this.submittedData = ko.observable({});
this.submissionMessages = ko.observableArray([]);

this.allUsers = ko.observableArray([]);

```

```

    this.allUsersDataProvider = ko.observable(new
ArrayDataProvider(this.allUsers(), { keyAttributes: 'email' }));

    this.hobbyRenderer = (context) => {
        return context.data && Array.isArray(context.data) ?
context.data.join(', ') : context.data;
    };

    this.goToNextStep = () => {
        if (this.stepValue() === 'step1') {
            const form1 = document.getElementById('personalInfoForm');
            if (form1 && !form1.checkValidity()) {
                this.step1Messages.push({
                    severity: 'error',
                    summary: 'Validation Error',
                    detail: 'Please fill in all required fields in Personal
Information.'
                });
                form1.reportValidity();
                return;
            }
            this.step1Messages([]);
            this.stepValue('step2');
        } else if (this.stepValue() === 'step2') {
            this.stepValue('step3');
        }
    };

    this.goToPreviousStep = () => {
        if (this.stepValue() === 'step2') {
            this.stepValue('step1');
        } else if (this.stepValue() === 'step3') {
            this.stepValue('step2');
        } else if (this.stepValue() === 'step4') {
            this.stepValue('step3');
        }
    };

    this.selectListener = (event) => {
        console.log("Files selected:", event.detail.files);
        this.profilePhoto(event.detail.files);
    };

    this.invalidListener = (event) => {
        console.log("Invalid file selected:", event.detail);
        this.submissionMessages.push({
            severity: 'warning',
            summary: 'Invalid File',

```

```

        detail: 'The selected file type is not allowed or too large.'
    });
};

this.handleSubmit = (event) => {
    if (event) event.preventDefault();

    if (this.stepValue() === 'step3') {
        const formData = {
            name: this.name(),
            email: this.email(),
            country: this.selectedCountry(),
            state: this.selectedState(),
            gender: this.gender(),
            hobbies: this.hobbies(),
            profilePhotoName: this.profilePhoto().length > 0 ?
this.profilePhoto()[0].name : 'None selected'
        };

        this.submittedData(formData);
        this.formSubmitted(true);

        this.allUsers.push(formData);

        this.allUsersDataProvider(new ArrayDataProvider(this.allUsers(), {
keyAttributes: 'email' }));
        console.log("allUsersDataProvider updated manually.");

        this.submissionMessages.push({
            severity: 'confirmation',
            summary: 'Submission Successful',
            detail: 'Your form has been submitted.'
        });

        const dialog = document.getElementById('submitDialog');
        if (dialog) dialog.open();
    }
};

this.resetForm = () => {
    this.name('');
    this.email('');
    this.selectedCountry('');
    this.selectedState('');
    this.gender('');
    this.hobbies([]);
    this.profilePhoto([]);
    this.formSubmitted(false);
};

```

```

    this.submittedData({});
    this.submissionMessages([]);
    this.step1Messages([]);
    this.stepValue('step1');
  };

  this.closeDialog = () => {
    const dialog = document.getElementById('submitDialog');
    if (dialog) dialog.close();
    this.stepValue('step4');
  };

  // FIX: Define tableElement *inside* the setTimeout callback's scope
  this.stepValue.subscribe((newValue) => {
    if (newValue === 'step4') {
      setTimeout(() => {
        // This declaration is now correctly inside the callback scope
        const tableElement = document.getElementById('userTable');
        if (tableElement) {
          //$(tableElement).ojTable('refresh');
          console.log("ojTable component refresh called.");
        } else {
          console.warn("User table element not found when trying to
refresh.");
        }
      }, 50);
    }
  });

  this.exportData = () => {
    const dataToExport = this.formSubmitted() ? this.submittedData() : {};
    if (Object.keys(dataToExport).length === 0) {
      this.submissionMessages.push({
        severity: 'info',
        summary: 'No Data to Export',
        detail: 'Please submit the form first to export data.'
      });
      return;
    }
    const jsonData = JSON.stringify(dataToExport, null, 2);
    const blob = new Blob([jsonData], { type: 'application/json' });
    const url = URL.createObjectURL(blob);
    const a = document.createElement('a');
    a.href = url;
    a.download = 'formData.json';
    document.body.appendChild(a);
    a.click();
    document.body.removeChild(a);
  };

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        URL.revokeObjectURL(url);
    };
}

return new FormPracticeViewModel();
});

```

6. now add code for formpractice.html

```

<div class="oj-hybrid-padding">
  <style>
    /* Styles for positioning the avatar */
    .avatar-container {
      position: absolute;
      top: 130px; /* Adjust as needed for vertical position */
      right: 100px; /* Adjust as needed for horizontal position */
      z-index: 10; /* Ensure it's above other content if overlapping */
    }
    /* Basic spacing for step content */
    .step-content {
      margin-top: 20px;
    }
    /* Margin for review items */
    .review-item {
      margin-bottom: 10px;
    }
    /* Style for oj-table */
    .my-user-table {
      width: 100%;
      height: 400px; /* Give the table a fixed height with scroll if many rows */
    }
  </style>

  <div class="oj-flex">
    <oj-train id="profileTrain"
      steps="[[trainSteps]]"
      selected-step="{{stepValue}}">
      <oj-train-step id="step1" label="Personal Information"></oj-train-step>
      <oj-train-step id="step2" label="User Preferences"></oj-train-step>
      <oj-train-step id="step3" label="Review and Submit"></oj-train-step>
      <oj-train-step id="step4" label="User Directory"></oj-train-step>
    </oj-train>
  </div>

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<div class="step-content" data-bind="if: stepValue() === 'step1'">
  <h2>Enter Your Personal Details</h2>
  <oj-messages messages="{{step1Messages}}"></oj-messages>
  <form id="personalInfoForm" class="oj-form-layout" max-columns="2"
direction="row" onsubmit="return false;">
    <oj-label for="nameInput">Enter name</oj-label>
    <oj-input-text id="nameInput" label-hint="Enter name" required="true"
value="{{name}}"></oj-input-text>

    <oj-label for="emailInput">Enter email</oj-label>
    <oj-input-text id="emailInput" label-hint="Enter email" required="true"
value="{{email}}"></oj-input-text>

    <oj-label for="countrySelect">Select Country</oj-label>
    <oj-select-single id="countrySelect" label-hint="Select Country"
      data="[[countryDataProvider]]"
      value="{{selectedCountry}}"
      required="true">
    </oj-select-single>

    <oj-label for="stateSelect">Select State</oj-label>
    <oj-select-single id="stateSelect" label-hint="Select State"
      data="[[stateDataProvider]]"
      value="{{selectedState}}"
      required="true"
      disabled="[[!isStateEnabled()]]">
    </oj-select-single>
  </form>
</div>

<div class="step-content" data-bind="if: stepValue() === 'step2'">
  <h2>Tell Us About Your Preferences</h2>
  <oj-form-layout>
    <div>
      <oj-label for="genderGroup">Gender</oj-label>
      <oj-radioset id="genderGroup" value="{{gender}}" required="true">
        <oj-option value="male">Male</oj-option>
        <oj-option value="female">Female</oj-option>
        <oj-option value="other">Other</oj-option>
      </oj-radioset>
    </div>

    <div>
      <oj-label for="hobbiesGroup">Hobbies</oj-label>
      <oj-checkboxset id="hobbiesGroup" value="{{hobbies}}">
        <oj-option value="reading">Reading</oj-option>
        <oj-option value="sports">Sports</oj-option>
        <oj-option value="music">Music</oj-option>
      </oj-checkboxset>
    </div>
  </oj-form-layout>
</div>

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        </oj-checkboxset>
    </div>

    <div>
        <oj-label for="profilePhoto">Profile Photo</oj-label>
        <oj-c-file-picker id="profilePhoto"
            accept="[[acceptArr]]"
            selection-mode="[[multipleStr]]"
            on-oj-select="[[selectListener]]"
            on-oj-invalid-select="[[invalidListener]]"
            disabled="[[isDisabled]]"
            label="Upload"></oj-c-file-picker>
        <span data-bind="text: profilePhoto().length > 0 ?
profilePhoto()[0].name : ''"></span>
    </div>
</oj-form-layout>
</div>

<div class="step-content" data-bind="if: stepValue() === 'step3'">
    <h2>Review Your Information</h2>
    <oj-messages messages="{{submissionMessages}}"></oj-messages>
    <div class="oj-form-layout">
        <oj-label-value label-edge="start" label-width="30%" class="review-
item">
            <oj-label slot="label" for="reviewName">Name</oj-label>
            <oj-input-text slot="value" id="reviewName" value="{{name}}"
readonly="true"></oj-input-text>
        </oj-label-value>
        <oj-label-value label-edge="start" label-width="30%" class="review-
item">
            <oj-label slot="label" for="reviewEmail">Email</oj-label>
            <oj-input-text slot="value" id="reviewEmail" value="{{email}}"
readonly="true"></oj-input-text>
        </oj-label-value>
        <oj-label-value label-edge="start" label-width="30%" class="review-
item">
            <oj-label slot="label" for="reviewCountry">Country</oj-label>
            <oj-input-text slot="value" id="reviewCountry"
value="{{selectedCountry}}" readonly="true"></oj-input-text>
        </oj-label-value>
        <oj-label-value label-edge="start" label-width="30%" class="review-
item">
            <oj-label slot="label" for="reviewState">State</oj-label>
            <oj-input-text slot="value" id="reviewState" value="{{selectedState}}"
readonly="true"></oj-input-text>
        </oj-label-value>
        <oj-label-value label-edge="start" label-width="30%" class="review-
item">

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        <oj-label slot="label" for="reviewGender">Gender</oj-label>
        <oj-input-text slot="value" id="reviewGender" value="{{gender}}"
readonly="true"></oj-input-text>
    </oj-label-value>
    <oj-label-value label-edge="start" label-width="30%" class="review-
item">
        <oj-label slot="label" for="reviewHobbies">Hobbies</oj-label>
        <oj-input-text slot="value" id="reviewHobbies"
value="{{hobbies().join(', ')}}" readonly="true"></oj-input-text>
    </oj-label-value>
    <oj-label-value label-edge="start" label-width="30%" class="review-
item">
        <oj-label slot="label" for="reviewPhoto">Profile Photo</oj-label>
        <oj-input-text slot="value" id="reviewPhoto"
value="{{profilePhoto().length > 0 ? profilePhoto()[0].name : 'None
selected'}}" readonly="true"></oj-input-text>
    </oj-label-value>
</div>

<oj-toolbar>
    <oj-button on-oj-action="[[handleSubmit]]">
        <span slot="startIcon" class="oj-ux-ico-check"></span> Submit
    </oj-button>
    <oj-button on-oj-action="[[resetForm]]">
        <span slot="startIcon" class="oj-ux-ico-undo"></span> Reset
    </oj-button>
    <oj-button on-oj-action="[[exportData]]">
        <span slot="startIcon" class="oj-ux-ico-export"></span> Export
    </oj-button>
</oj-toolbar>
</div>

<div class="step-content" data-bind="if: stepValue() === 'step4'">
    <h2>Submitted User Directory</h2>
    <pre data-bind="text: JSON.stringify(allUsers(), null, 2)"></pre>

    <oj-table
        id="userTable"
        aria-label="Submitted Users Table"
        data="[[allUsersDataProvider]]"
        columns='[
            {"headerText": "Name", "field": "name"},
            {"headerText": "Email", "field": "email"},
            {"headerText": "Country", "field": "country"},
            {"headerText": "State", "field": "state"},
            {"headerText": "Gender", "field": "gender"},

            {"headerText": "Photo", "field": "profilePhotoName"}
        ]'
    >

```

```

    ]'
    scroll-policy="loadMoreOnScroll"
    class="oj-flex-item my-user-table"
    style="width:100%; height:400px;">
  </oj-table>
</div>

<div class="avatar-container">
  <oj-avatar id="userAvatar"
    role="img"
    aria-label="User avatar"
    size="lg"
    initials="[[avatarInitials]]">
  </oj-avatar>
</div>

<div class="oj-flex oj-sm-justify-content-center oj-sm-margin-top-4">
  <oj-button data-bind="click: goToPreviousStep, visible: stepValue() !==
'step1'">Back</oj-button>
  <oj-button data-bind="click: goToNextStep, visible: stepValue() !==
'step4'" class="oj-sm-margin-start-2">Next</oj-button>
</div>

<oj-dialog id="submitDialog" dialog-title="Submission Successful"
modality="modeless">
  <div slot="body">
    <p>Your form was submitted successfully!</p>
  </div>
  <div slot="footer">
    <oj-button on-oj-action="[[closeDialog]]">OK</oj-button>
  </div>
</oj-dialog>

<div class="oj-sm-margin-top-4" data-bind="if: formSubmitted">
  <h3>Submitted Data (JSON)</h3>
  <pre data-bind="text: JSON.stringify(submittedData(), null, 2)"></pre>
</div>
</div>

```

7. Run ojet serve on command terminal
8. Navigate to the Form Practice page in the browse.
9. Enter the required fields and click on next. The first two pages remains the same as per previous practice but there are some advancements at the step 3 page. Here you will notice one more step at the top “user directory” as train step 4 and there is a toolbar

at the bottom that contains the submit, reset and export icons. also, at the bottom, there is a back and next option at the review and submit page

ORACLE App Name john.hancock@oracle.com

Dashboard Incidents Customers About Form Practice

1 Personal Information 2 User Preferences 3 Review and Submit 4 User Directory

Review Your Information

Name richa jain
Email demo@gmail.com
Country IN
State DL
Gender male
Hobbies reading
Profile Photo photo.jpg

✓ Submit ↶ Reset 📄 Export

Back Next

On clicking at export, the file gets exported and downloads in json format.

When you click on submit option , you will see the successful entry confirmation dialog box along with the submitted data in json format. (it is displayed there just to ensure, the data is correctly binding)

Review Your Information

✓ Confirmation

Submission Successful
Your form has been submitted.

Name richa jain
Email
Country
State
Gender
Hobbies
Profile Photo

✓ Submit ↶ Reset 📄 Export

Back Next

Submitted Data (JSON)

```
{
  "name": "richa jain",
  "email": "richa1238@gmail.com",
  "country": "IN",
  "state": "PB",
  "gender": "female".
}
```

When you click on Ok button to close this dialog box, you will move to the next train step4 “user directory”

[Dashboard](#) [Incidents](#) [Customers](#) [About](#) [Form Practice](#)

1 Personal Information

2 User Preferences

3 Review and Submit

4 User Directory

RJ

Submitted User Directory

```
[
  {
    "name": "richa jain",
    "email": "richa123@gmail.com",
    "country": "IN",
    "state": "PB",
    "gender": "female",
    "hobbies": [
      "reading"
    ],
    "profilePhotoName": "photo.jpg"
  }
]
```

Name ↕	Email ↕	Country ↕	State ↕	Gender ↕	Photo ↕
richa jain	richa123@gmail.com	IN	PB	female	photo.jpg

Here you will see that data is automatically entered into the table, which is ojet control component. You can make the multiple entries to this but email should be unique. Otherwise, it will not accept new entries.

Click on back button and reset the form, to make new entry. And then again submit it. Now you will see two entries in the table.

1 Personal Information

2 User Preferences

3 Review and Submit

4 User Directory

A

Submitted User Directory

```
[
  {
    "name": "richa jain",
    "email": "richa123@gmail.com",
    "country": "IN",
    "state": "PB",
    "gender": "female",
    "hobbies": [
      "reading"
    ],
    "profilePhotoName": "photo.jpg"
  },
  {
    "name": "admin",
    "email": "admin@india.com",
    "country": "IN",
    "state": "DL",
    "gender": "male",
    "hobbies": [
      "sports"
    ],
    "profilePhotoName": "None selected"
  }
]
```

Name ↕	Email ↕	Country ↕	State ↕	Gender ↕	Photo ↕
richa jain	richa123@gmail.com	IN	PB	female	photo.jpg
admin	admin@india.com	IN	DL	male	None selected

Congratulations !! Here you have completed this practice, where you have introduced the ojet control component “**oj-table**” with some advanced features.